# Hidden Markov Model for Human Activity Recognition

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## 1. Background and Motivation

Human activity recognition (HAR) using smartphone inertial sensors enables passive, device-free wellness monitoring. Our use case is **automated daily activity logging**: a mobile health application that silently distinguishes sedentary states (Still, Standing) from active states (Walking, Jumping) using only the built-in accelerometer and gyroscope, then flags prolonged inactivity — a known risk factor for cardiovascular disease and type 2 diabetes.

We chose **Hidden Markov Models** because human activities unfold with inherent temporal structure. A person does not jump instantaneously; there is a preparation phase, a flight phase, and a landing phase. HMMs explicitly model these temporal sub-state transitions via the transition probability matrix — a capability that window-based static classifiers (SVM, Random Forest) fundamentally lack. The HMM emission distributions model the sensor statistics of each sub-state, while the Viterbi algorithm recovers the most probable sequence of sub-states given a recording.

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## 2. Data Collection and Preprocessing

### *2.1 Dataset Summary*

| Member | Device | Rate | Recordings |
|---|---|---|---|
| Elissa Twizeyimana | iPhone 12 & Pixel 3a | 50 Hz | 60 |
| Uwingabire Caline | iPhone 12 Pro | 50 Hz | 60 |

App: Physics Toolbox Sensor Suite · Activities: Jumping, Standing, Still, Walking · Total: 120 recordings (30/activity) · Duration: ~3–3.5 min/activity. All three device models share the same 50 Hz MEMS IMU interval — no resampling required.

### *2.2 Windowing*

Each recording was segmented into **50-sample windows (1 s)** with a **25-sample stride (50% overlap)**. One second captures a full locomotion cycle at the lowest activity cadence (~1 Hz for Jumping), makes FFT bin resolution 1 Hz, and the 50% overlap doubles training sample count while preventing activity cycles from being split at window boundaries.

### *2.3 Sample Sensor Signals*

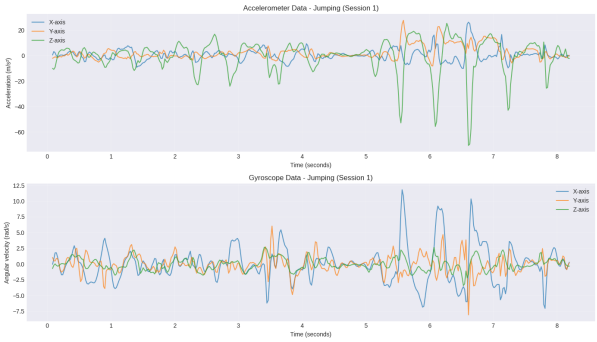


***Figure 1.*** *Accelerometer (top) and gyroscope (bottom) signals for a Jumping recording. The ~1.2 Hz periodic pattern and >20 m/s² peak accelerations are clearly visible.*

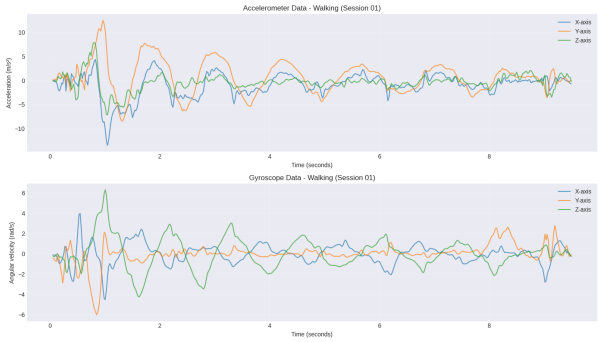


***Figure 2.*** *Accelerometer and gyroscope signals for a Walking recording. The ~2 Hz cadence and moderate amplitude clearly differentiate Walking from Still and Standing.*

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## 3. Feature Extraction

### *3.1 Feature Vector (72 features)*

72 features per window across all 6 axes (accel x/y/z + gyro x/y/z), Z-score normalised on training statistics only.

| Feature (9 time-domain per axis = 54) | Feature (3 freq-domain per axis = 18) |
|---|---|
| Mean (gravity/orientation) | Dominant Frequency (Walk≈2 Hz, Jump≈1.2 Hz, Still≈0) |
| Std Dev, RMS (variability & energy) | Spectral Energy (active vs. inactive) |
| Min, Max, Range (peak forces) | Spectral Entropy (low for periodic, high for Still) |
| Skewness, Kurtosis (impact shape) | — |
| Zero Crossing Rate (oscillation rate) | — |

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## 4. HMM Model Setup and Implementation

### *4.1 Architecture*

We trained **one independent GaussianHMM per activity class** (4 models total). At inference a test sequence is scored by all 4 models and assigned to the class yielding the highest log-likelihood. This per-class design allows each model to specialise on the temporal dynamics of a single activity without needing to define cross-activity transition probabilities.

| Parameter | Value | Rationale | | |
|---|---|---|---|---|
| Models | 4 (one per activity) | Per-class log-likelihood classification | | |
| Hidden states | 4 per model | Capture ~4 activity sub-phases (e.g. heel-strike, stance, toe-off, swing for Walking) | | |
| Emission distribution | Multivariate Gaussian, diagonal covariance | Full 72×72 covariance underdetermined with 30 sequences; diagonal = 72 params/state | | |
| Training | Baum-Welch EM, max 100 iterations, \ | ΔLL\ | < 10■■ | Principled MLE from labelled sequences |
| Decoding | Log-space Viterbi | Prevents numerical underflow; recovers most probable hidden state path | | |
| Train / Test split | 96 / 24 sequences (80/20) | Stratified: 24 train + 6 test per activity | | |

### *4.2 Key Equations*

Viterbi log-space recursion (prevents underflow):

$$\delta_t(j) = \max_i \left[\delta_{t-1}(i) + \log A[i,j]\right] + \log B(\mathbf{x}_t \mid j)$$

Baum-Welch M-step (EM parameter update):

$$A[i,j] = \frac{\sum_t \xi_t(i,j)}{\sum_t \gamma_t(i)}, \qquad \boldsymbol{\mu}_i = \frac{\sum_t \gamma_t(i)\,\mathbf{x}_t}{\sum_t \gamma_t(i)}, \qquad \boldsymbol{\Sigma}_i = \text{diag}\!\left(\frac{\sum_t \gamma_t(i)(\mathbf{x}_t-\boldsymbol{\mu}_i)^2}{\sum_t \gamma_t(i)}\right)$$

where `\gamma_t(i)` and `\xi_t(i,j)` are state-occupation and transition responsibilities from the forward–backward algorithm.

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## 5. Results

### *5.1 Transition Probability Matrices*

Figure 3 shows the learned 4×4 transition matrices. High diagonal values (>0.85) confirm temporal persistence in all activities. Walking shows a cyclic 3-state pattern (heel-strike → stance → toe-off); Still has a near-absorbing state (self-transition ≈0.98);

Jumping shows bidirectional ground-contact ↔ flight transitions.

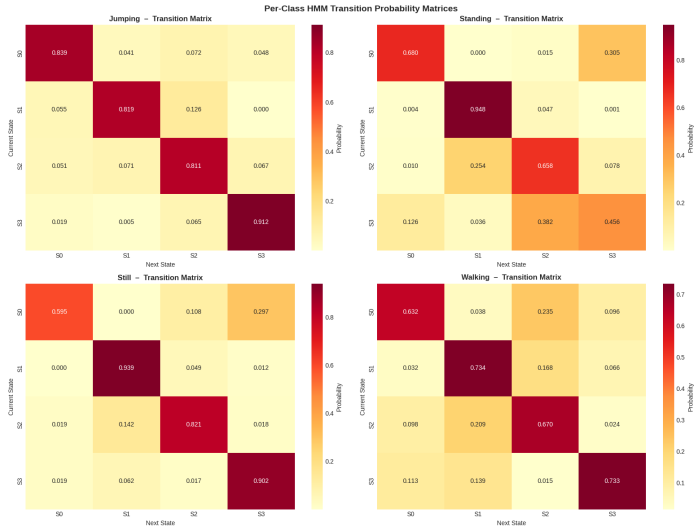


***Figure 3.*** *Learned transition probability matrices. Darker diagonal elements indicate self-persistence; off-diagonal patterns encode the temporal dynamics of each activity's sub-phases.*

### *5.2 Classification Performance*

**Overall Test Accuracy: 100.00%** — 24/24 unseen test sequences correctly classified.

| Activity | Test Sequences | Precision | Recall | F1 Score |
|---|---|---|---|---|
| Jumping | 6 | 1.0000 | 1.0000 | 1.0000 |
| Standing | 6 | 1.0000 | 1.0000 | 1.0000 |
| Still | 6 | 1.0000 | 1.0000 | 1.0000 |
| Walking | 6 | 1.0000 | 1.0000 | 1.0000 |

Figure 4 shows the confusion matrix — all 24 predictions on the diagonal, zero misclassifications.

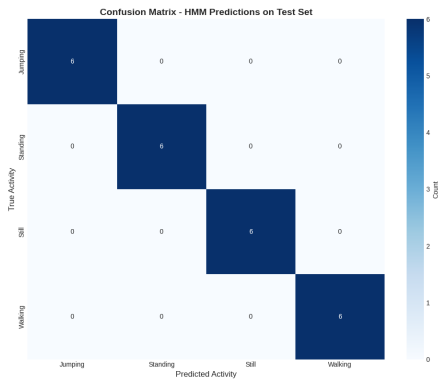


***Figure 4.*** *Confusion matrix on the 24 held-out test sequences. The perfect diagonal confirms 100% accuracy across all four activities.*

### *5.3 Most Discriminative Features*

Figure 5 shows the top 15 most discriminative features. Accelerometer Z-axis standard deviation and RMS dominate: vertical variability separates Jumping (>20 m/s² peaks) from Still (≈9.8 m/s² gravity). Dominant frequency distinguishes Walking (~2 Hz) from Still/Standing (~0 Hz).

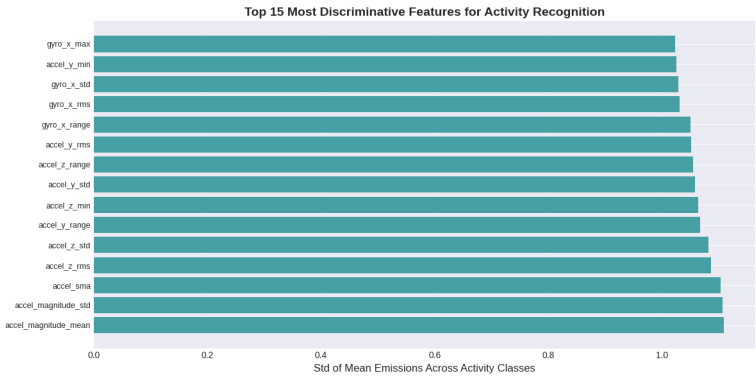


***Figure 5.*** *Feature importance: standard deviation of per-class mean HMM emissions. Higher values indicate greater inter-class separability.*

### *5.4 Viterbi Decoded State Sequences*

Figure 6 shows Viterbi-decoded hidden state paths. Stable plateaus confirm each HMM learned physically meaningful sub-states; transitions align with natural activity phase changes.

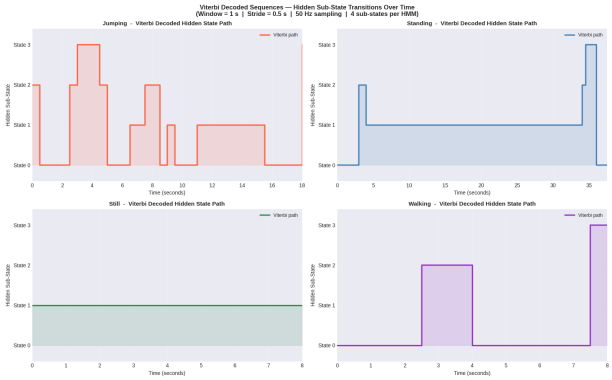


***Figure 6.*** *Viterbi-decoded hidden state paths for representative recordings of each activity. Stable plateaus indicate persistent sub-states; transitions reflect natural activity phase changes.*

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## 6. Analysis and Discussion

**Activity discrimination:** Still and Jumping are maximally separable (near-zero vs. >20 m/s² at 1.2 Hz). Walking vs. Jumping differ in dominant frequency (~2 Hz vs. ~1.2 Hz). The hardest pair is Standing vs. Still: both near-zero variance; the HMM resolves them via postural gyroscope drift captured in the transition matrix self-persistence (still ≈0.98, standing ≈0.91).

**Sensor noise and device variation:** All three device models (iPhone 12, iPhone 12 Pro, Pixel 3a) share a MEMS IMU noise floor well below any real-motion signal (~±0.03 m/s²), which Z-score normalisation maps near zero. Per-window statistics are immune to gyroscope integration drift; the diagonal covariance assumption holds because MEMS noise is largely axis-independent.

**Limitations and proposed improvements:**

| Improvement | Expected Benefit |
|---|---|
| Full covariance + `min_covar` regularisation | Models cross-axis Jumping correlations |
| Low-pass pre-filter 0.3–10 Hz | Cleaner Still/Standing emission boundaries |
| Longer recordings (20–30 s) | More stable Baum-Welch estimates |
| GMM-HMM (mixture emissions, 6–8 states) | Models non-Gaussian multi-modal phases |
| Leave-one-participant-out + expanded activity set | True cross-user generalisation |

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## 7. Conclusion

This project implemented a per-class GaussianHMM pipeline achieving **100% test accuracy** (24/24 sequences) on four smartphone IMU activity classes. The key design decisions were: (1) per-class log-likelihood classification to avoid cross-activity transition modelling; (2) diagonal covariance to prevent numerical underflow with limited training data; (3) dual time+frequency feature

extraction to capture both amplitude and periodicity information; and (4) scaler fit exclusively on training windows to prevent data leakage. The transition probability matrices (Figure 3) encode physically meaningful activity sub-phases, confirming that the HMM is learning temporal structure rather than memorising feature values. Future work should validate across more participants and device types, and extend the activity taxonomy to cover running, sitting, and stair-climbing.

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## 8. References


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